Evaluation of the Shielding Design of the LINAC Bunker at the New INMP Building, AERE, BAEC, Bangladesh

A thesis report submitted to the Department of Medical Physics and Biomedical Engineering, Faculty of Post Graduate, Gono Bishwabidyalay in partial fulfillment of requirement for the degree of Master of Science in Medical Physics and Biomedical Engineering.

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Declaration

It is hereby declared that this thesis has been conducted ethically. The data and the results presented are the genuine and obtained by me during the working period. This thesis report or any part of it has not been submitted elsewhere for the award of any degree or diploma.

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Signature of the candidate



DEDICATED TO

MY PARENTS

MY UNCLE (Dr. MIR LATIAR HOSSAIN)

AND

MY TEACHERS

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LIST OF SYMBOLS

A_0	Beam area at the first scattering surface (m^2)
A_1	Area of wall that can be seen from maze door (m^2)
A_z	Cross-sectional area of maze inner entry (m^2)
B_L	Barrier of Leakage
B_{pri}	Primary Barrier
B_{sca}	Barrier of Scatter
μ	Linear Attenuation Coefficient
a	is the scatter fraction
dscat	distance source to scatter patient
dsec	distance from patient to point of interest
F	Field area at mid-depth of the patient at isocenter (cm2)
Gy	Gray
mSv	Milli-Sievert
nTVL	Number of Tenth Value Layer
P	Allowed dose rate per week outside the barrier
T	Occupancy factor
U	Use factor
W	Work load

LIST OF ABBREVIATIONS

3DCRT	3 Dimensional Conformal Radiotherapy
AERE	Atomic Energy Research Establishment
BAEC	Bangladesh Atomic Energy Commission
HVL	Half-value layer
IMRT	Intensity modulated radiation therap
INMP	Institute of Nuclear Medical Physics
LINAC	Linear accelerator
MF	Modulation Factor
MV	Mega Voltage
NCRP	National Council on Radiation Protection
QC	Quality control
SRS	Safety Report Series
SRS	Stereotactic Radio Surgery
TBI	Total Body Irradiation
TVL	Ten-value layer
VMAT	Volumetric Modulated Arc Therapy

ABSTRACT

Introduction: This study showed that the LINAC machine for radiotherapy technique requiresthe shielding calculation of the LINAC Bunker. This LINAC machine is used to treat cancerpatients in the photon of a maximum of 18 MV. As the particle energies are in the MV range, specialmeasurements and calculations must need to be taken into account for designing the LINACbunker. Otherwise, the dose limit for the public and radiation personnel will exceed the limit ofthe protocol. This study has been performed at the Institute of Nuclear Medical Physics(INMP), AERE, Saver, Dhaka. The design of these LINAC rooms is different comparedwith the general projects cause two LINAC machines are installed alongside each other. These several sources of ionizing radiation are too much harmful if the shielding calculationis considered with a little ignorance. The purpose of the study is to attenuate radiation from LINAC Machine in where existing thickness is evaluated by calculated thickness.

Methods and Materials: This study provides essential guidelines for proper facilityplanning, Room design feature and shielding design, radiation protection, and work practice.Calculations were done assuming concrete as the shielding design. We proposed a shieldingcalculation model that decouples the concepts of occupancy factor, workload,use factor, andtarget dose when determining primary and secondary barrier thickness. All the equation forshielding calculation is according to NCRP-151,NCRP-49& SRS-47.

Results and Discussion: The patient numbers and occupancy factors are considered as taken variables in thiscalculation IMRT & 3DCRT shielding calculation. Maximally 18MV linear accelerator isconsidered for the photon beam here. We get the values of the calculations: workload-450Gy/week; WQA-112.5Gy/week; Total Workload, W-563Gy/week and we evaluated the exiting value and calculated value.

Conclusion: As physicists,our aim for us is to consider and calculate the shielding to protect the public and radiation personnel. This paper has discussed what measures should have taken to protect occupational workers and the public from LINAC radiation hazards. These LINAC are located inside the bunkers. The main purpose of the bunker is to absorb the radiation produced by the LINAC, such that the people outside the bunker do not affect by the ionizing radiation.

CHAPTER 1: INTRODUCTION

1.1 General Introduction

Linear accelerators (LINAC) are used to produce a high-energy beam of radiation by almost all radiotherapy units. Cancer cells are treated with high-energy beams of radiation. These beams are a type of ionizing radiation and healthy human cells can be damaged also. Safety protocols must be maintained to limit radiation exposure to hospital workers and normal people. Linac must be kept in a shielded room that inhibits the radiation outside the radiotherapy facility to achieve a safe exposure limit for the non-patients.

During the first 50 years of radiotherapy, technological progress was relatively slow but over the past decade the way that therapeutic radiation treatments for tumors are delivered has become much more precise. Technological advancements have led to treatment machines that can deliver very small and precise beams of radiation, and are highly conformal, such as intensity-modulated radiotherapy (IMRT), three-dimensional Conformal radiotherapy (3DCRT), volumetric modulated arc therapy(VMAT) tomotherapy, stereotactic radiosurgery(SRS), etc.[1]

A radiotherapy department is a special structural building constructed to protect people from radiation hazards. Radiotherapy treatment involves a doctor, medical physicists, nurses, and technicians regularly and occasionally involves the patient and the general public. Radiation is harmful to health, although they are useful in some important cases for mankind. For such use, one has to ensure protection against radiation. The primary aim of radiological protection is to provide an appropriate standard of protection for people without unduly limiting the beneficial practices giving rise to radiation exposure. All those who are concerned with radiological protection have to make value judgments about the relative importance of different kinds of risk and about balancing risks and benefits. As a consequence of the penetrating nature of the X-ray beam, they need to be sited in a shielded bunker to reduce the external dose rates and annual doses to meet the constraints imposed by national and international regulations.

Structural shielding design in medical radiotherapy installations aims to limit radiation exposures to members of the public and employees to an acceptable level, i.e. to reduce the effective dose from a linear accelerator (linac) to a point outside the radiotherapy bunker as low as reasonably achievable. Shielding design is particularly concerned with attenuation of the primary beam and secondary radiation in the form of head leakage, and patient and wall scatter. Thus, finding the optimum barrier thickness is an essential requirement for the safety of radiotherapy facilities.[2]

This study shows that the LINAC machine for radiotherapy technique requires the shielding calculation of the LINAC Bunker. We proposed a shielding calculation model that decouples the concepts of occupancy factor, workload, use factor, and target dose when determining primary and secondary barrier thickness. All the equation for shielding calculation is used according to NCRP-151,NCRP-49 & SRS- 47.

The main purpose of the bunker is to absorb the radiation produced by the LINAC, such that the people outside the bunker do not affect by the ionizing radiation.

1.2 Aims of Study

The main goal of this study is to evaluate the shielding design of the linac bunker and find outtheadequate thickness of the protective primary barrier and secondary barriers.As the project of LINAC bunker is under construction. So, it is a revision work.

Objectives of the study are-

General Aims:

- ❖ To ensurethe LINAC bunker shielding
- ❖ To ensure the public safety and health professionals

Specific Aims:

* To calculate theoretically the required thickness of all barriers of the LINAC bunker.
* To evaluated the thickness between the theoretical result and the existing thickness of under constructed LINAC bunker.

1.3 Scope of Study

This project has been done in the Department of medical physics and Biomedical Engineering, Gono Bishwabidyalay. On the other hand,this work has been performed at the Institute of Nuclear Medical Physics (INMP), AERE, Saver, Dhaka. This work and all the associated calculations have been done according to NCRP-151,NCRP-49, and SRS-47.

1.4 Literature Reviews

For the successful completion of this thesis work different literature has been reviewed from different journals, websites, and other relevant sources. These reviews revealed the similarity and dissimilarities with the thesis work which is discussed below:

In an article of M. F. Uddin, M. M. Ahasan,R.KhatunS. Akter.et. al.[3] it has been demonstrated ona Shielding Design Basis and its Calculation of High Energy Medical Linac Installed in Bangladesh Atomic Energy Commission, Bangladesh.This study was described onHigh energy medical linac from Varian Medical Systems of model ClinaciX installed in Bangladesh Atomic Energy Commission with dual photon energies (15 MV and 6 MV) and five electron energies upto 18 MeV. This machine is used to treat cancer patients either in electron or photon mode depending on the position of the malignant tissue. As the particle energies are in the MeV range, special measures are taken into account for designing a linac bunker. In this study, maximum photon energy is considered for shielding design. The other two particles i.e., electron and neutron are not considered due to their low penetration depth. As physicists, the aim for us is to consider and calculate the shielding so that to protect the people and the stuff. Protection is required against three types of radiation such as primary radiation (from the patient), scattered radiation (from the patient), and leakage radiation (from the linac head). To save the radiation worker from these three types of radiation, primary wall and secondary wall thickness, primary wall width, and roof thickness are calculated. This paper reveals what measures have been taken to protect occupational workers and the public from linac radiation hazards.

A Research paper byCliffordDugal,B.Eng et.al [4] has been described on the application of Monte Carlo to Linac Bunker Shielding Design. The purpose of this study Medical linear accelerators (linacs) are frequently used to treat cancer patients. These linacs are located inside bunkers. The bunker's main purpose is to contain the radiation produced by the linac, such that the people outside the bunker do not suffer from radiation-induced illnesses. The current design method for bunkers relies on simplified equations. These equations do not always result in bunkers being built that match the designed protection criteria. This work aims to look at the use of Monte Carlo for bunker shielding calculations, as Monte Carlo more closely models radiationphysics. It is hoped that this better modeling will lead to more accurate bunker designs.

In this work, an 18 MV Siemens PRIMUS linac was modeled in MCNP, and promising comparisons to theory, measurements, and calculations were made.Areachers paper by James Rjiken, Scott Crow et.al.[5]has been described as Linear accelerator bunker shielding for stereotactic radiotherapy.Shielding protocols such as NCRP 151, IAEA SRS 47, and IPEM 47 are commonly used for the design of radiotherapy facilities. Some work has been accomplished in updating the basic formula with the advent of IMRT but little consideration has been made for unflattened beams and stereotactic techniques apart from for facilities housing devices like the CyberKnife. The inevitable scenario of a stereotactic-only linear accelerator was considered in this study. The necessary shielding requirements were determined based on stereotactic data from a year's worth of treatments from one clinic as well as further measurements of leakage, scatter, and use factors. These values were compared to recommendations in the literature. While tenth-value layer amounts, workload, and barrier widths could be kept at the status quo, major changes could be made to the parameters of leakage, scatter, and use factors while still maintaining safety. Some differences could also be seen in the use of IMRT factors. Current shielding protocols were found to inadequately describe a methodology for the shielding of a stereotactic-only radiotherapy linac bunker, producing overly conservative wall thicknesses which are in disagreement with the principles of ICRP.

In an article by James Rjiken, Madhava Bhat, Scott Crow et.al.[6]it has been described on Conservatism in linear accelerator bunker shielding. Conservatism in the shielding of linear accelerator bunkers is engrained in the methodology of international protocols and guidelines. However, the degree to which this cautious and prudent approach is necessary should be judged against the International Committee of Radiation Protection's principles of exposure justification and optimization. Radiation survey data from 75 concrete barriers were aggregated and compared to exposure predictions from three popular protocols to assess any conservatism in factors used to calculate scatter, leakage, and beam penetration. These findings, in addition to a list of common conservative practices, were then used to tally the possible fiscal impact of an over-conservative approach to linear accelerator bunker shielding. While primary beam penetration was accurately predicted, stated conservatisms in scatter and leakage were found to be largely misplaced. An estimated total factor of conservatism calculated from a tally was found to agree with the literature values of radiotherapist occupational exposure. This factor amounted to a cost increase of 43% for a single bunker if all conservative assumptions weremade. There are aspects of linear accelerator shielding design that are overly conservative, beyond what is

justifiable by the International Committee of Radiation Protection. Some adjustments to international protocol methodology may be required.

A paperbyJRijken, S Towns, and B Healy [7], was described the need to update NCRP 151 data for 10 MV linear accelerator bunker shielding based on new measurements and Monte Carlo simulations. The study of the performance of the Linear accelerator bunker shielding protocols such as NCRP 151 haspreviously been tested against a large sample of measured data from real bunkers and machines but differences in per-energy concrete penetration (TVLs) for 10 MV have not yet been resolved. These differences are likely due to historical beam data and can potentially result in the over-exposure of radiation workers andthe public. This study examines a cohort of clinical linac bunker survey measurements and compares them to popular shielding protocols. Differences were investigated using contemporary beam data for both Monte Carlo simulation and analytical equations. For primary barriers, NCRP 151 underestimates the dose rate through concrete by on average a factor of 2 with secondary barriers and maze entrance doses having much better agreement. The use of contemporary beam data in Monte Carlo simulation and an analytical equation yielded TVL values much closer to the measured values compared to NCRP 151. The TVL data in NCRP 151 is outdated and needs to be updated based on the energy spectra of modern linear accelerators. Until then, physicists should use the TVL values shown in this study instead.

A research paper by Andrew Mc Morrow, St. Vincets's Private Hospital, Elm Park [8],has been performed on Radiation Shielding and Bunker Design.The performed of this study was theThis project aimed to develop a computerized algorithm that could be used to accurately assess the protection required (in terms of dose rates) in the design of Radiotherapy bunkers. This included primary, scattered photon beams and neutron scattered beams. There are commercial computerized models available (mainly Monte Carlo based), but they are expensive and smaller institutes do not have readily available access to these models. Most radiotherapy departments rely on various publications on the subject of bunker design such as and use empirical formulae to estimate the protection required. The formulae tend to be incorporated in spreadsheet format but may become cumbersome. However, they must be revisited each time for accuracy and modernization, as often the LINAC 1 (Linear Accelerator) specification and the design criteria (dose constraint) change with time. Given the above, creating an automated model /solution to shielding design calculations for high-energy radiation was the task. For the above, there were existing calculated and measured dose rates for the bunkers which facilitated comparisons of results. Once the accuracy was determined, other bunker designs were assessed.

A journal paper byJulia Jank, Gabriele Kragl, and Dietmar Georg[9],was described on Impact of a flattening filter-free linear accelerator on structural shielding design.The present study aimed to assess the effects of a flattening filter-free medical accelerator on the structural shielding demands of a treatment vault of a medical linear accelerator. We tried to answer the question, of to what extent the required thickness of the shielding barriers can be reduced if instead of the standard flattened photon beams unflattened ones are used. We chose both an experimental as well as a theoretical approach. On the one hand, we measured photon dose rates at protected places outside the treatment room and compared the obtained results for flattened and unflattened beams. We complied with international guidelines for adequate treatment vault design

and calculated the shielding barriers according to the therein given specifications. Measurements were performed with an Elekta Precise™ linac providing nominal photon energies of 6 and 10 MV. This machine underwent already earlier some modifications to be able to operate both with and without a flattening filter.

An article byDominikHenzen, Daniel Schmidhalter, and Claudia Christina Zanella[10], has been described on the Evaluation of clinically applied treatment beams concerning bunker shielding parameters for a Cyberknife M6. The study was Compared to a conventional linear accelerator, the Cyberknife (CK) is a unique system concerning radiation protection shielding and the variety and number of non-coplanar beams are two key components regarding this aspect. In this work, a framework to assess the direction distribution and modulation factor (MF) of clinically applied treatment beams of a CyberKnife M6 is developed. Database filtering options allow studying the influence of different parameters such as collimator types, treatment sites, or different bunker sizes. A distribution of monitor units (MU) is generated by projecting treatment beams onto the walls, floor, and ceiling of the CyberKnife bunker. This distribution is found to be highly heterogeneous and depends, among other parameters, on the bunker size. For our bunker design, 10%-13% of the MUs are delivered to the right and left walls, each.

Research by Glenn P Glasgow [11] has described Structural Shielding Design and Evaluation for Megavoltage X- and Gamma-Ray Radiotherapy Facilities. The purpose of radiation shielding is to limit radiation exposure to members of the public and employees to an acceptable level. This Report presents recommendations and technical information related to the design and installation of structural shielding for megavoltage x- and gamma-ray radiotherapy facilities. In addition, the use of barriers constructed with composite materials has become commonplace.

This thesis was performed on the Shielding Design of the LINAC Bunker for 18MV.The entire method follows NCRP-151,NCRP-49,and SRS-47 guidelines.

CHAPTER 2: THEORITICAL BACKGROUND

2.1 Primary Barrier

- Primary barriers are the portions of treatment room walls, floor, or ceiling that may be irradiated directly by the primary beam of LINAC.
- In this case, the north, south, and ceiling walls are considered as Primary barrier

2.2 Secondary Barrier

- Secondary barriers are all portions of treatment room walls that can not be irradiated directly by the primary beam, or not in the direction of the primary beam.
- In this case, the east and west walls are considered as Secondary barriers.

2.3. Recommended dose limits

Table 1: Summary of recommended/legal effective dose limits and design of effective dose limits for different protocols

Dose limit	BAERA	IAEA	USA	UK
Occupational exposure dose limit	20 mSv per year averaged over 5 years and 50 mSv in any single year	20 mSv per year averaged over 5 years and 50 mSv in any single year	Implied annual limit of 10 mSv, cumulative dose of age ×10mSv and 50 mSv in any single year	20 mSv in a year or 100 mSv in 5 consecutive years and 50 mSv in single year
Design limit for occupational exposure	-	-	Fraction of 10 mSv annually	6 mSv in a year IDR is 7.5 μSv/h
Public dose limit	1 mSv in a year	1 mSv in a year	Infrequently, 5 mSv annually, and continually, 1 mSv annually	1 mSv in a year
Design limit for public area	-	-	1 mSv annually IDR is 20 μSv/h	0.3 mSv in a year IDR is <7.5 μSv/h

2.4 Shielding Materials

Table 2: Material used for shielding calculation

Material	Density (kg^{-3})	Comment
Breeze blocks	1100-1400	Up to 500kV with supplementary lead or steel shielding
Clay bricks	1600	
Earth fill	1600	Useful if bunker below ground level
Concrete (poured or blocks)	2350	Density can vary with mineral content
High-density concrete (poured or blocks)	3800-4600	Density-dependent on iron ore content
Leadite®/Verishield®	3700-4000	Interlocking blocks
Steel	7900	Normally used as supplemental shielding on existing rooms
Lead	11340	

2.5 Primary Barrier Width

For bunkers containing linear accelerators where the radiation beam is confined to a a rotational plane through the isocentre (sometimes termed 'C-arm accelerators'), the primary barriers will be limited in width to reduce concrete volume and cost. The width of the barrier will primarily be set by the extent of the beam at the barrier distance using the diagonalofthelargest field size, commonly 400 mm ×√2 = 570 mm at the isocentredistance.This barrier intrudes into the treatment room, where 300 mm is added at the distance of the inner wall provided by the secondary shielding and extends outside the room, where 300 mm is added at the external wall. The width of the primary wall barrier will need to be maintained forthe primary barrier in the roof of the bunker if the exterior is accessible.

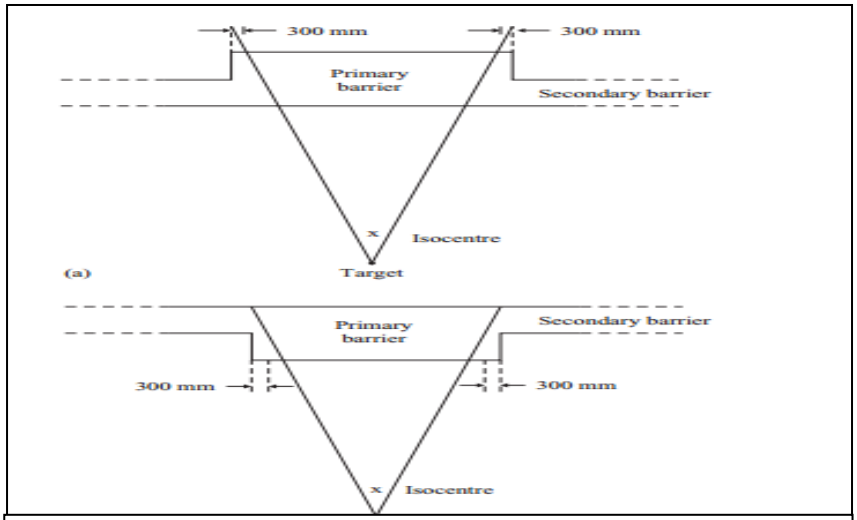


Figure 1 (a): Primary barrier external to treatment room. (b) Primary barrier intruding into treatment room.

2.6 Linear Accelerators

Linear accelerators are now the most common treatment units in radiotherapy departments incancer centers and units. They produce high-energy X-ray and electron beams. Most models of linear accelerators produce two or three beams of X-rays and typically five electron beams, each with a different energy to facilitate a range of patient treatments. As a consequence of the penetrating nature of the X-ray beam, theyneed to be sited in a shielded bunker to reduce the external dose rates and annual dosesto meet the constraints imposed by national legislation. The two common arrangements for safely housing a linear accelerator are shown in the plan in Figures (a) and (b). Both layouts show the increased thickness of shielding (primary shielding) needed to attenuate the X-ray beam when it falls directly on a wall after passing through the patient. An increased thickness in shielding will also be required in the roof of the bunker in the area directly irradiated by the x-ray beam and in the floor if the treatment units are sited above occupied areas. The layouts also show the thinner shielding (secondary shielding) required in the other walls to attenuate the lower energy leakage radiation from the treatment head of the linear accelerator and the radiation scattered by the patient and from the directly irradiated walls.

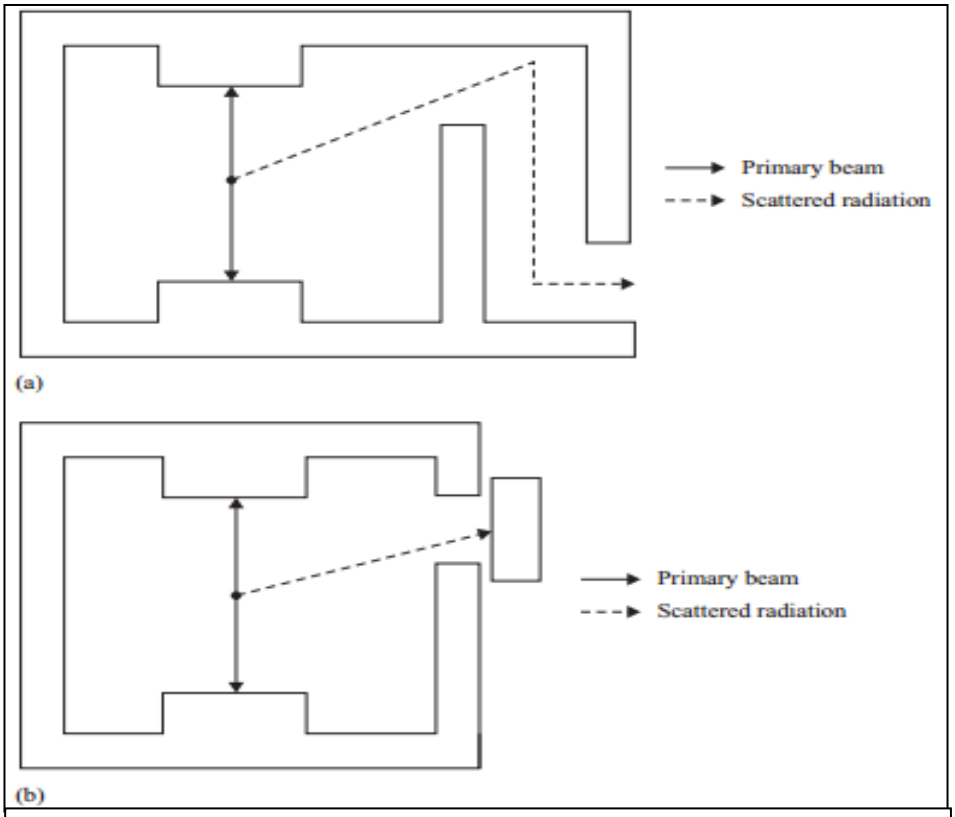


Figure 2: Shows that primary barrier and secondary barrier

2.7 The Checklist of Requirements for a Treatment Facility with a Linear Accelerator:

Access

For equipment installation and maintenance
For calibration and quality assurance equipment
For patients on trolleys
Dosimeter cable route

Equipment services

Stabilized electrical supply
Chilled cooling water
Compressed air (for some linear accelerators)
UPS for computers
IT network connections

Health and Safety

Shielding
Maze and/or shielding door
Radiation warning signs
Radiation warning lights
Emergency stop buttons

Door/barrier interlock
Last person out button
Treatment room
Laser alignment devices
In-room monitors
Lighting level control
Mount for gating camera
Connections to IT network
Wash basin(s) and water supply
Treatment room storage
Immobilization devices
Applicators
Electron beam cut-outs
Control area
Connections to IT network
Dosimeter cable route to the treatment room
Control area storage
Patient notes and radiographs
Clinical procedures
Safety procedures
Quality assurance procedures and records
Equipment fault log
Dosimetry charts
Patient communication
CCTV
Voice intercommunication
General
Patient changing facilities
Patient waiting area
Other storage requirements
Equipment performance and maintenance records
Quality assurance equipment
Spare parts and test equipment

2.8 Kilovoltage units

The Cancer Treatment Facilities: Planning and Design Manual (DH 2011) describessuperficial/orthovoltage treatment units as optional, but most centers have one suchunit. This might be an orthovoltage unit operating at X-ray energies up to 300 kV ora superficial unitoperating at two or three X-ray energies up to a maximum of150 kV for the treatment of superficial and near-surface conditions[12].

2.9 Shielding Terminology

2.9.1 Controlled Area

The glossary of the BSS defines a controlled area as any area in which specific protection measures and safety provisions are or could be required for:

1. Controlling normal exposures or preventing the spread of contamination during normal working conditions.
2. Preventing or limiting the extent of potential exposure.[13]

The employees who work in controlled areas have significantpotential forexposure to radiation in the course of their assignments or are directly responsible for or involved with the useand control of radiation. Generally, these employees have training in radiation management and are subject to routine personalmonitoring. NCRP recommends an annual limit for *E* for these individuals of 50 mSv y–1 with the cumulative *E* not to exceed the product of10 mSv and the worker's age in years (exclusive of medical and natural background radiation) (NCRP, 1993). That notwithstanding,NCRP (1993) recommends that for the design of new facilities, *E should* be a fraction of the 10 mSv y–1 implied by the cumulativeeffective dose limit. Another consideration is that a pregnant radiation worker *should not* be exposed to levels that result greater than the monthly equivalentdose (*H*T) limit of 0.5 mSv to the worker's embryo or fetus (NCRP, 1993). To achieve both recommendations, this Report recommends a fraction of one-half of that *E* value, or 5 mSv y–1, and a weekly shielding design goal (*P*) of 0.1 mSv dose equivalent (*H*) (*i.e.*, an annual *H* value of 5 mSv) for controlled areas. The *P* value adopted in this Report would allow pregnant radiation workers continued access to their work areas.Recommendation for Controlled Areas:Shielding design goal (*P*) (in dose equivalent): 0.1 mSv week–1(5 mSv y–1).

2.9.2 Uncontrolled Areas

Uncontrolled areas are those occupied by individuals such as patients, visitors to the facility (e.g., patient visitors, delivery service representatives, and consultants), and employees who do not work routinely with or around radiation sources. Areas adjacent to, but not part of, the radiotherapy facility are also uncontrolled areas. Based on ICRP (1991) and NCRP (1993) recommendations for the annual limit of effective dose to a member of the public, shielding designs shall limit exposure of all individuals in uncontrolled areas to an effective dose that does not exceed 1 mSv y^{-1}. After a review of the application of the guidance in NCRP (1993) to medical radiation facilities, NCRP concluded that a suitable source control for shielding individuals in uncontrolled areas in or near medical radiation facilities is an effective dose of 1 mSv in any year (NCRP, 2004). This recommendation can be achieved for the medical radiation facilities covered in this Report with a weekly shielding design goal (P) of 0.02 mSv dose equivalent (H) (i.e., an annual H value of 1 mSv) for uncontrolled areas. Recommendation for Uncontrolled Areas: Shielding design goal (P) (in dose equivalent): 0.02 mSv $week^{-1}$ (1 mSv y^{-1}) [2].

2.9.3 P - Design dose

- Also referred to as 'limit dose' per specified period

- Usually based on 20 mSv per year for occupationally exposed persons, and 1 mSv for public

- Can apply additional constraint e.g. 0.3 (to account for the fact that a person can be irradiated from multiple sources at the same time)

- Occupational dose only to be used in controlled areas i.e. only for radiographers, physicists, and radiation oncologists

2.9.4 Workload

The workload (*W*) for radiotherapy equipment covered in this Report is the time integral of the absorbed-dose rate determined at the depth of the maximum absorbed dose, 1 m from the source. The most common period over which *W* is specified is one week.The units for W areGy week–1 and conversion to a workload W2 at a distance d2 different than 1m would be W2 = W (1 m)2 / (d2)2. The value for W is usually specified as the absorbed dose from photons delivered to the isocenter in a week and is selected for each accelerator based on its projected use. This is usually estimated from the average number of patients (or fields) treated in a week and the absorbed dose delivered per patient (or field). It should also include an estimate of the average weekly absorbed dose delivered during quality control checks, calibrations, or other physics measurements. Treatments on modern clinical accelerators often involve using low- and high-energy X-ray beams and electron beams of various energies. For dual-energy machines, the workload at the higher energy will usually determine the shielding requirement. However, in some situations, to determine the required barrier thicknesses for both primary and secondary radiations it may be necessary to consider separately the workloads for each x-ray beam quality. The workload for electron beam operation can be disregarded, except for shielding accelerators with electron beam-only operation, such as dedicated intraoperative facilities. Modern radiotherapy facilities often employ techniques, such as intensity-modulated radiation therapy (IMRT), that build up an absorbed-dose distribution in the target volume through the accumulation of multiple beamlets (small area beams). These treatment delivery methods can lead to leakage-radiation workloads that are significantly greater than the total absorbed dose at the isocenter, and this has led various authors to create a workload efficiency factor as well as to decouple the primary and secondary beam workloads[14].

2.9.5 Use Factor

The use factor (U) is the fraction of a primary-beam workload that is directed toward a givenprimary barrier. The value for U will depend on the type of radiation installation. For example, atraditional facility with a beam that rotates about an isocenter will usually have a symmetricdistribution of gantry treatment angles and these will be predominately in the four primaryangles(0, 90, 180, and 270 degrees). However, a facility that is used for total-body irradiation (TBI) withthe patient at an extended distance will have a large use factor for the direction of the TBItreatments. And facilities that perform a large number of tangential breast treatments will havesignificant use factors for oblique room angles[15].

2.9.6 Occupancy Factor

Occupancy factor T is a factor with which workload is multiplied to account for the degree of occupancy of the area in question. Whenever possible, the local situation should be assessed before determining the occupancy factor to be used[16].

2.9.7 HVL and TVL

The half-value layer (HVL or $X_{1/2}$) is defined as the thickness of the attenuator that attenuates the photon beam intensity to 50% of its original value.

$x_{1/2}$=HVL=(ln2)/μ

The tenth-value layer (HVL or $X_{1/10}$) is defined as the thickness of the attenuator that attenuates the photon beam intensity to 10% of its original value[17].

$x_{1/10}$=HVL=(ln2) /μ

2.9.8 Maze

To reduce the radiation dose near the entrance a restricted access passageway leading to the room may be incorporated in the design. This passageway is termed the maze. Ldealle\y this should be as long and with as small a cross-section as possible. The minimum width may be determined by the dimensions of the treatment unit to be delivered by this route or by access for a hospital bed. A maze ensures that photon radiation can only exit the room after scattering and attenuated. A maze reduces the need for a heavy shielding door[18].

CHAPTER 3: MATERIALS AND METHODS

3.1 Materials

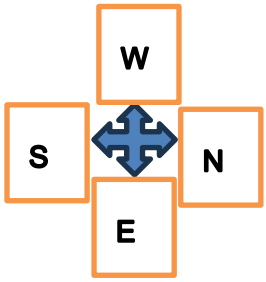
- Protocals:

  -Protection and Measurements NCRP report no-151

  - NCRP report no-49

  - SRS-47

- Concrete-2350kgm^-3
- Building Design of the Institute of Nuclear Medical Physics (INMP) W



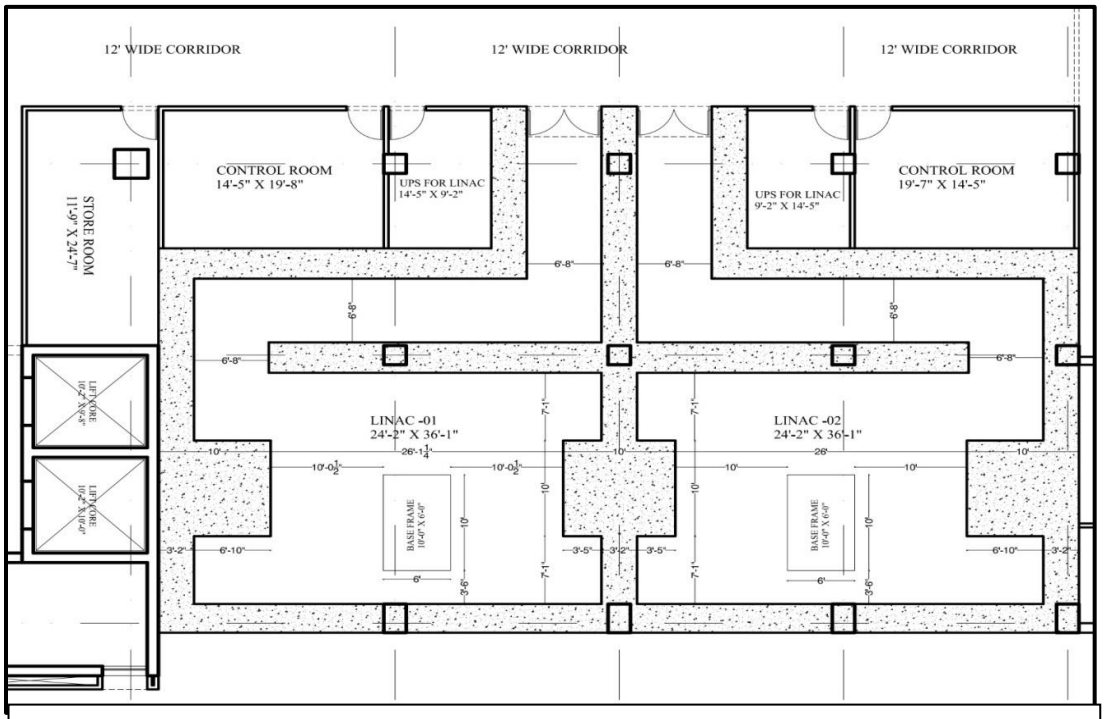


Figure 3: Top View of 2 Bunkers for LINACs

3.3 Calculation of Primary Barrier Thickness

The equations used to calculate the Primary Barrier were-

$B_{pri} = \frac{Pd^2}{WUT}$…….…… (1)

The required number (n) of TVLs is calculated as

$nTVL = -Log(B_{pri})$……(2)

And the barrier thickness is given by:

Thickness= TVL_1 + (n-1) TVL_e……(3)

Table 3: (Ceiling) Primary radiation

Parameters	Symbol	Value
Design limit at the point of interest	*P (Sv/week)*	0.00002
Distance from the isocenter to the point of interest	*d (m)*	5.70
Workload	*W*	563
Use factor	*U*	0.25
Occupancy factor	*T*	1
Barrier transmission factor	B_{pri}	0.00000462
Number of TVL in ordinary concrete	*N*	5.34
First TVL	TVL_1 *(m)*	0.45
Equivalent TVL	TVL_e *(m)*	0.43
Barrier thickness	T_{pri} *(m)*	**2.3142**

Table 4: (North wall) Primary radiation

Parameters	Symbol	Value
Design limit at the point of interest	*P (Sv/week)*	0.00002
Distance from the isocenter to the point of interest	*d (m)*	3.96
Workload	*W*	563
Use factor	*U*	0.25
Occupancy factor	*T*	0.5
Barrier transmission factor	B_{pri}	0.00000447
Number of TVL in ordinary concrete	*N*	5.35
First TVL	TVL_1 *(m)*	0.45
Equivalent TVL	TVL_e *(m)*	0.43
Barrier thickness	T_{pri} *(m)*	**2.3205**

Table 5: (South Wall) Primary radiation

Parameters	Symbol	Value
Design limit at the point of interest	*P (Sv/week)*	0.00002
Distance from the isocenter to the point of interest	*d (m)*	3.96
Workload	*W*	563
Use factor	*U*	0.25
Occupancy factor	*T*	0.5
Barrier transmission factor	B_{pri}	0.00000447
Number of TVL in ordinary concrete	*N*	5.35
First TVL	TVL_1 *(m)*	0.45
Equivalent TVL	TVL_e *(m)*	0.43

Barrier thickness	T_{pri} *(m)*	**2.3205**

3.4 Calculation of Secondary Barrier

The equations used to calculate Leakage Radiation was-

$B_L = \frac{Pd_s^2}{WUT}$……(4)

The equation used to calculate Scattered Radiation was-

$B_P = \frac{Pd_{sca}^2 d_{sec}^2}{aWT(\frac{F}{400})}$……(5)

Table 6: (East Wall) Leakage radiation (Eq-4)

Parameters	Symbol	Value
Design limit at the point of interest	*P (Sv/week)*	0.0004
Distance from the isocenter to the point of interest	d_s *(m)*	3.68
Workload	*W*	1941
Occupancy factor	*T*	0.025
Barrier transmission factor	B_L	0.11183604
Number of TVL in ordinary concrete	*N*	0.95
First TVL	TVL_1 *(m)*	0.45
Equivalent TVL	TVL_e *(m)*	0.43
Barrier thickness	S_c *(m)*	**0.4291**

Table 7: (East Wall) Scattered radiation (Eq.-5)

Parameters	Symbol	Value
Design limit at the point of interest	*P (Sv/week)*	0.0004
Distance from the isocenter to the point of interest	d_{sec} *(m)*	3.68
Distance from target to the isocenter	d_{scat} *(m)*	1
Workload *(*30 Patient per day× 5 days a week× 3 Gy per Patient=450*Gy/week)*	*W*	563

Occupancy factor	*T*	0.025
Scattered fraction defined at 1 m	*a* (10^0)	0.0142
Field Size	*F/400*	0.04
Barrier transmission factor	B_L	0.00679286
Number of TVL in ordinary concrete	*n*	2.17
First TVL	TVL_1 *(m)*	0.45
Barrier thickness	S_c *(m)*	**0.9522**

The difference between the scatter and leakage barrier thickness (0.9522–0.429= 0.5231m) is larger than one TVL, therefore, the larger thickness has taken.

Table 8: (West Wall) Leakage Radiation (Eq.-4)

Parameters	**Symbol**	**Value**
Design limit at the point of interest	*P (Sv/week)*	0.0004
Distance from the isocenter to the point of interest	d_s *(m)*	5.72
Workload	*W*	1941
Occupancy factor	*T*	1
Barrier transmission factor	B_L	0.00673210
Number of TVL in ordinary concrete	*N*	2.17
First TVL	TVL_1 *(m)*	0.45
Equivalent TVL	TVL_e *(m)*	0.43
Barrier thickness	S_c *(m)*	**0.9539**

Table 9: (West Wall) Scattered radiation (Eq.-5)

Parameters	Symbol	Value
Design limit at the point of interest	*P (Sv/week)*	0.0004
Distance from the isocenter to the point of interest	d_{sec} *(m)*	5.72
Distance from target to the isocenter	d_{scat} *(m)*	1
Workload	*W*	563
Occupancy factor	*T*	1
Scattered fraction	*a* (10^0)	0.0142
Field Size	*F/400*	0.04
Barrier transmission factor	B_P	0.0000890
Number of TVL in ordinary concrete	*N*	3.39
First TVL	TVL_1 *(m)*	0.45
Barrier thickness	S_c *(m)*	**1.477**

The difference between the scatter and leakage barrier thickness (1.477– 0.9539 = 0.5231m) is larger than one TVL. Therefore, the larger thickness has taken.

3.5 Dose at the maze door

The equations used to calculate the Dose at the ma maze door,

$$H_G = fH_S + H_{LS} + H_{ps} + H_{LT} \quad \text{.....(6)}$$

The radiation scattered to the maze door,

$$\boldsymbol{H_S = \frac{WU_G\alpha_0 A_0 \alpha_z A_z}{(d_h d_r d_z)\text{^}2}} \quad \text{.....(7)}$$

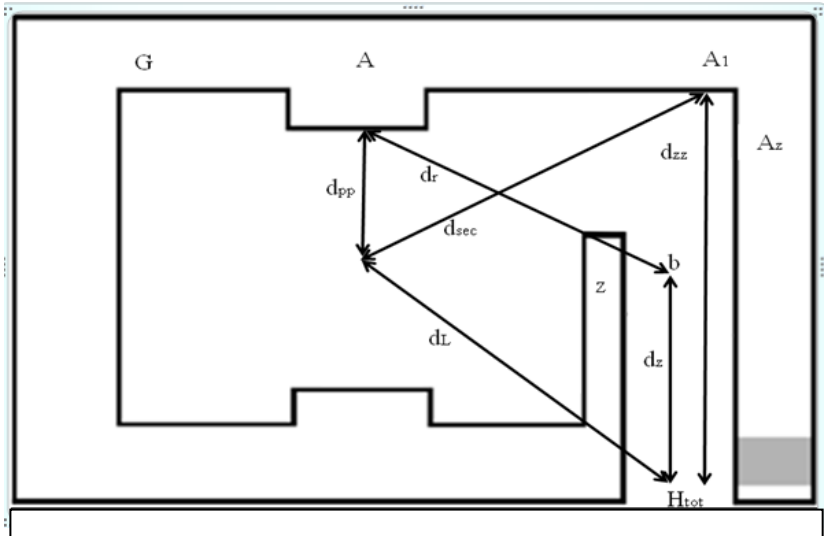


Figure 4: Schematic diagram of proposed LINAC room for calculation capture gamma and neutron dose at the maze

Table 10: Dose arising from wall scatter (Figure-4)

Parameters	Symbol	Value
Workload	*W (Gy/Week)*	563
Use factor	U_G	0.25
Reflection coefficient at the first scattering surface	α_0	0.0052
Beam area at the first scattering surface	A_0 (m^2)	2.5
The wall reflection coefficient at	α_z	0.022
Area of the inner maze entrance	A_z (m^2)	17
The perpendicular distance from the isocenter to the wall,	d_h *(m)*	4.00
Distance from beam center at the first reflection, past the edge of the inner maze wall, to Point b on the midline of the maze	d_r *(m)*	11.33
Centerline distance along the maze from Point b to the maze door	d_z *(m)*	10.74
Dose equivalent per week at the maze door due to scattering of the primary beam from the Wall	**Hs(Gy/Week)**	**0.000000000218**

Head-leakage radiation:

$$H_{LS} = \frac{L_f W U_G \alpha_1 A_1}{(d_{sec} d_{zz})^2} \text{.....(8)}$$

Table 11: Dose arising from head leakage (Figure-4)

Parameters	Symbol	Value
The fraction of the dose due to head leakage at 1.0 m from the source	L_f	0.001
Workload	*W (Gy/Week)*	1941
Barrier use factor	U_G	0.25
Reflection coefficient from wall G	α_1	0.0064
Area of Wall G that can be seen from the maze door	A_1 (m^2)	11.15
Distance from the radiation source to the maze centerline	d_{sec} *(m)*	10.74
Centerline distance along the maze	d_{zz}*(m)*	8.90
Head-Leakage Wall-Scattered Radiation Component	H_{LS}(Gy/Week	**0.000001285149**

Patient scatter component (H_{ps}):

$$H_{PS} = \frac{W U_0 a \left(\frac{F}{400}\right)(\alpha_1 A_1)}{(d_{sca} x d_{sec} x d_{ZZ})^2} \text{.....(9)}$$

Table 12: Dose arising from patient scatter (Figure-4)

Parameters	Symbol	Value
Workload	*W (Gy/Week)*	563
Use factor	U_g	0.25
The scatter function at the patient	α_1	0.00139
The field area incident on the patient (40x40 cm^2)	*F /400*	0.04
Source to isocenter distance	d_{sca} *(m)*	1.00

Distance from the patient to Wall G at the maze centerline	$d_{sec}(m)$	8.24
Centerline distance along the maze	d_{zz} *(m)*	10.74
Area of the G wall that can be seen from maze	A_1 (m^2)	11.15
The wall reflection coefficients	α	0.022
Patient –Scattered Radiation Component	H_{ps}(Gy/Week)	**0.000001285149**

Head leakage through the maze wall (H_{LT})

$$\boldsymbol{H_{LT} = \frac{L_f W U_G 10\text{^}(\frac{t}{TVL})}{(d_l - d)^2}} \ldots\ldots\ldots\

$$H_{cg} = Wxh_{\varphi} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (13)$$

Table 14: Capture gamma dose at the maze door

Parameters	Symbol	Value
The apparent neutron source strength	Q_N *(Neutrons/Gy)*	950000000000.000000
Distance from the isocenter to Location	d_1 *(m)*	7.000000
Distance from Location A to the door	d_2 *(m)*	9.700000
The surface area of the treatment room	*Sr(m^2)*	526.000000
Workload	*W (Gy/week)*	1941
Transmission factor for neutrons that penetrate the head shielding	β	1.000000
Dose equivalent (sievert) to the total neutron fluence at location A	*K Sv.m^2 per unit neutron fluence*	0.000000
Tenth-value distance	*TVD(m)*	5.400000
The total neutron fluence at the inner maze point A	ϕ_A*(m^{-2}G^-2*	3468716260
The equivalent weekly capture gamma dose at the maze door	*Hcg(Sv/Week)*	**0.000007**

3.7 Neutron dose at the maze door

At the maze entrance, the neutron dose equivalent is usually the dominant component for high-energy accelerators above 10 MV. The neutron dose at any point in the maze depends on several factors, including the distance from the inner maze point (A_{zz}) to the isocentre (d_1), the surface area (S) of the treatment room, the inner maze entrance cross-sectional area (A_r) and the cross-sectional area of the maze (S_1). It is also a function of the energy, the gantry angle, and the field size of the photon beam.

Total Dose Equivalent at the Maze Door:

$$H_w = H_{tot} + Hc_g + H_n \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (14)$$

The neutron dose calculation at the maze entrance (with an additional bend) is given by the following equation

$$H_{n,D} = 2.4 \times 10^{-15} \Phi_A \sqrt{\frac{S_0}{S_1}} [1.64 \times 10 + 10^{-\left(\frac{d_2}{1.9}\right)} + 10^{-\left(\frac{d_2}{TVD}\right)} \times W_L \ldots\ldots\ldots\ldots (15)$$

Therefore, the equivalent weekly dose at the maze door due to neutrons is given by the following equation:

$$Hn = W\ x\ H_n, D \ldots\ldots\ldots\

CHAPTER 4: RESULT AND DISCUSSIONS

4.1 Summary of barrier thickness

Linac bunkers were designed to prevent unintentional exposure of staff, patients and the public. The minimum distance to the occupied area from a shielded wall is assumed to be 0.3 m. Our designed thickness values are more than theoretical values. This is due to practical experience and consultation with other linac facilities shielding experts.

Table 16: Difference Between Constructed Value and Calculated Value

Parameter	Constructed Value	Calculated Value	Difference
Unit	Constructed Thickness(Th) m		
Ceiling	3.048	2.314	0.733
North	3.048	2.321	0.727
South	3.048	2.321	0.727
East	0.965	0.952	0.013
West	1.930	1.470	0.453

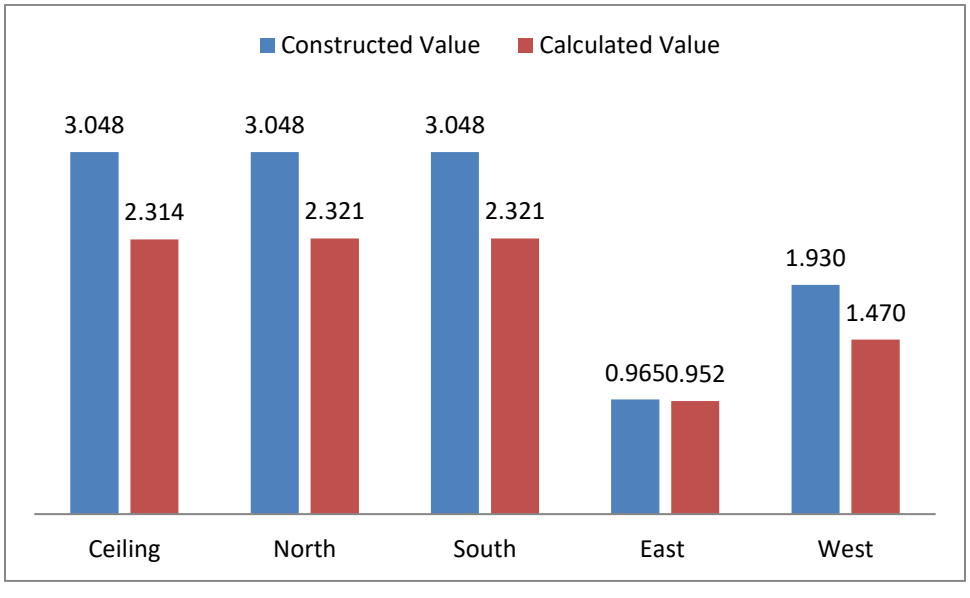


Figure 6: Difference Between Constructed Value and Calculated Value

4.2 Discussions

4.2.1 Bunker Shielding Calculation

Values are calculated from empirical formulas following the international guidelines presented in tabular form. It is assumed that the roof and primary layer thickness should be the same. But in the calculation, roof thickness is slightly different from the primary layer due to the variation of occupancy and use factor values. Physically thin wall made of steel plate or lead blocks may create ground shine beneath the wall, which add an extra dose rate transmitted through the wall. In that case, additional shielding on the floor at the foot of the wall may be necessary to reduce the ground shine to an acceptable dose rate. Our primary and secondary barrier thickness for 18 MV. This large thickness minimizes the probability of ground shine. Concrete density is considered to be 2.35 g cm-3. From practical experience, sometimes the concrete mixing ratio is not maintained properly. So, our designed thickness values are more than theoretical values. In designing a room, use a long maze has been chosen instead of a direct shield door. The length of the maze for energies higher than 10 MV is not done in this calculation.

CHAPTER 5: CONCLUSION

5.1 Conclusion

The shielding requirements for a LINAC facility vary from energy to energy. LINAC bunkers are designed to prevent unintentional exposure of staff, patients, and the public.Our designed thickness values are more than theoretical values. This is due to practical experience and consultation with other LINAC facilities shielding experts. The scattered photons may exit through a door in the room unless that door is hidden by a maze. Our long maze prevents the scattered photons to escape from the bunker by a significant amount. At high energies, neutrons scattering are multiple times compared with that of photons.

The thicknesses of primary and secondary radiation barriers for an 18 MV LINAC bunker were calculated following the guidelines of NCRP 151,NCRP 49,and SRS 47. Careful planning and shielding design helps to optimize protection and safe costs. Shielding design and calculations are complex and must be performed by a qualified radiation expert based on sound assumptions. All shielding must be checked by an independent expert. This work has demonstrated complete guidelines for shielding calculation and radiation for a high-energy LINAC.

CHAPTER 6: REFERENCES

6.1 REFERENCES

1.Shieding calculation of vlumetric modulated ARC therapy (VMAT).NazimAman,Dpt. of Medical Physics and Biomedical Engineering , Gono Bishwabidyalay, Saver, Dhaka;

2.Shielding calculation and verification for 15MV medical Linear accelerator treatment facilities.Mohammad Ullah, dept. of Medical Physics and Biomedical Engineering ,Gono Bishwabidyalay.

3.Shielding Design Basis and its Calculation of High Energy Medical Linac Installed in Bangladesh Atomic Energy Commission, Bangladesh,10.13189/ujmsj.2017.050202

4.Application of Monte Carlo to linac bunker shielding design,DOI.org (Crossref)

5.Linear accelerator bunker shielding for stereotactic radiotherapy,10.1088/1361-6560/ab4916

6.Conservatism in linear accelerator bunker shielding,10.1007/s13246-019-00782-1

7.The need to update NCRP 151 data for 10 MV linear accelerator bunker shielding based on new measurements and Monte Carlo simulations,10.1088/1361-6498/ac2e0b

8.Radiation Shielding and Bunker Design,10.26717/BJSTR.2019.18.003106

9.Impact of a flattening filter free linear accelerator on structural shielding design, 10.1016/j.zemedi.2013.05.002

10. Evaluation of clinically applied treatment beams with respect to bunker shielding parameters for a Cyberknife M610.1002/acm2.12215

11.Structural shielding design and evaluation for megavoltage X- and gamma-ray radiotherapy facilities: recommendations of the National Council on Radiation Protection and Measurements,978-0-929600-87-1

12.Design and Shielding of Radiotherapy Treatment Facilities,IPEM Report 75, 2nd Edition,Patrick Horton Royal Surrey County Hospital, Guildford, UK ,David Eaton Mount Vernon Cancer Centre, London, UK

13.safety Reports Series, No-47**,** Radiation Protection in the Design of Radiotherapy Facilities.

14.Recommendations of the NATIONAL COUNCIL ON RADIATION PROTECTION AND MEASUREMENTS,(NCRP-151)*December 31, 2005*

15.Institute of Physics and Engineering in Medicine, Medical and Dental Guidance Notes, IPEM, Yourk (2002)

16.Verification and Justification of Value Shielding Calculated for 3DCRT and IMRT, Mohammmed Shaheen Miah,Dpt. Of Medical Physics and Biomedical Engineering , Gono Bishwabidyalay, Saver, Dhaka;

17.Radiation Oncology Physics: A Handbook for Teachers and Students, E.B. Podgorsk

18.Shielding Calculation of Intensity Modulated Radiation Therapy (IMRT), TahmidaAkter,Dpt. Of Medical Physics and Biomedical Engineering , Gono Bishwabidyalay, Saver, Dhaka;

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